

Dylan Dai

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EDUCATION

University of Waterloo

Bachelor of Computer Science (Honours)

Waterloo, Ontario

WORK EXPERIENCE

Celestra

Founding Member of Technical Staff

September 2025 – May 2026

San Francisco, CA

- **Employee #1** at seed stage startup backed by **General Catalyst** to enhance pharmaceutical market research
- Built automated multi-agent systems achieving frontier performance on MLE-Bench, peaking at **rank 2 globally**
- Worked with **3 top-10** global pharma firms to build production analytics powered by our autonomous AI agent
- Built **entire** post-training suite for concurrently active autonomous ML agents, with live training and checkpoint deployment across distributed GPU fleets
- Built and trained a **synthetic digital twin** suite for physicians and patients, to simulate market research at scale
- Automated pharma data analysis across **800K+** HCPs and **1.2B** rows, replacing manual data science workflows
- Mined **400+ gigabytes** of Reddit and Facebook data to uncover rare-disease patient insights for market research

Cohere

Software Engineer Intern

May 2025 – August 2025

Toronto, ON

- Saved **\$100,000+** **monthly** and **35%** in runtime by adding batching to all company-wide AI model calls
- Reduced evaluation effort by **30%** by building a tool to access all Cohere's AI model benchmark statistics
- Saved **20%** in GPU runtime by adding cost tracking and aggregation for all company-wide AI model calls
- Reduced server load for storing AI model queries by **90%** from implementing decision trees for item indexing

Cohere

Data Engineer Intern

August 2024 – April 2025

Toronto, ON

- Managed coding [datasets](#) used to train state of the art machine learning model **Command-A**

PUBLICATIONS

Investigating Component Contributions in Multi-Agent ML Systems

ICML 2026

Junsung Kim, Ilija Mireskandari, Seungwan Son, Yifan Zhou, Khizer Shahid, **Dylan Dai**

- Ran **4,000+** ablation experiments on ML engineering agents; showed iterative feedback beats architectural complexity.
- Built **K-live**, a contamination-free benchmark of **25** active competitions for leakage-resistant ML agent evaluation.

PROJECTS

AI Dataset Undersampler | NumPy, TypeScript, Three.js, Scikit-learn

- Diversifies AI model training datasets by **30%** by building a tool to analyze and filter similar data using k-means
- Visualized data flow by embedding data then using Principal Component Analysis for vector compression

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, C, Bash, HTML/CSS

Frameworks & Tools: PyTorch, NumPy, Pandas, Git, AsyncIO, GCP, AWS, TensorFlow, Transformers, PostgreSQL

Interests: Fashion, Digital Art, Tetris, Puzzle Games, Thrifting, Guitar, Rubik's Cubes, Travelling

AWARDS AND ACHIEVEMENTS

Canadian Computing Olympiad Bronze Medalist | Placed **14th** out of 10,000+ participants

National ranked Tetris player | **Top 50** global on tetris.com

Hackathons | Won **\$2,000+** in three hackathons totalling **1500+** participants